8.2 Single-Volume Component, SNGLVOL

A single-volume component is indicated by SNGLVOL for Word 2 on card CCC0000. The junction connection code determines the placement of the volume within the system. More than one junction may be connected to an inlet or outlet. If an end has no junctions, that end is considered a closed end. Normally, only a branch has more than one junction connected to a volume end. For major edits, minor edits, and plot variables, the volume in the single-volume component is numbered as CCC010000.

8.2.1 Single-Volume X-Coordinate Volume Data, CCC0101-0109

This card (or cards) is required for a single-volume component. The nine words can be entered on one or more cards, and the card numbers need not be consecutive.

W1(R) Area (m^2, ft^2).

W2(R) Length (m, ft).

W3(R) Volume (m^3, ft^3). The program requires that the volume equals the area times the length (W3 = W1•W2). At least two of the three quantities, W1, W2, and W3, must be nonzero. If one of the quantities is 0, it will be computed from the other two. If none of the words are 0, the volume must equal the x-direction area times the x-direction length within a relative error of 0.000001. The same relative error check is done for the y- and z-directions.

W4(R) Azimuthal angle (degrees). The absolute value of this angle must be ≤ 360 degrees and is defined as a positional quantity. This quantity is not used in the calculation but is specified for possible automated drawing of nodalization diagrams.

W5(R) Inclination angle (degrees). The absolute value of this angle must be ≤ 90 degrees. The angle 0 degrees is horizontal; and positive angles have an upward inclination, i.e., the inlet is at the lowest elevation. This angle is used in the interphase drag calculation.

W6(R) Elevation change (m, ft). A positive value is an increase in elevation. The absolute value of this quantity must be less than or equal to the length. If the vertical angle orientation is 0, this quantity must be 0. If the vertical angle is nonzero, this quantity must also be nonzero and have the same sign. When the absolute value of the elevation angle determined by the ratio of the elevation change (this Word 6) and the length (Word 2) is less than or equal to 45 degrees, the horizontal flow regime map is used. When the ratio is greater than 45 degrees, the vertical flow regime map is used.

W7(R) Wall roughness (m, ft).

W8(R) Hydraulic diameter (m, ft). This should be computed from $4.0 \bullet \frac{\text{area}}{\text{wetted perimeter}}$. If 0, the hydraulic diameter is computed from $2.0 \bullet \left(\frac{\text{area}}{\pi}\right)^{0.5}$. A check is made to ensure the pipe roughness is less than half the hydraulic diameter. See Word 1 for area.

W9(I) Volume control flags. This word has the format tlpvbfe. It is not necessary to input leading zeros. Volume flags consist of scalar oriented and coordinate direction oriented flags. Only one value for a scalar oriented flag is entered per volume but up to three coordinate oriented flags can be entered for a volume, one for each coordinate direction. At present, the f flag is the only coordinate direction oriented flag. This word enters the scalar oriented flags and the x-coordinate flag.

t-flag The digit t specifies whether the thermal front tracking model is to be used; t = 0 specifies that the front tracking model is not to be used for the volume, and t = 1 specifies that the front tracking model is to be used for the volume.

l-flag The digit l specifies whether the mixture level tracking model is to be used; l = 0 specifies that the level model not be used for the volume, and l = 1 specifies that the level model be used for the volume.

p-flag The digit p specifies whether the water packing scheme is to be used. p = 0 specifies that the water packing scheme is to be used for the volume, and p = 1 specifies that the water packing scheme is not to be used for the volume.

v-flag The digit v specifies whether the vertical stratification model is to be used. v = 0 specifies that the vertical stratification model is to be used for the volume, and v = 1 specifies that the vertical stratification model is not to be used for the volume.

b-flag The digit b specifies the interphase friction that is used. b = 0 means that the pipe interphase friction model will be applied, b = 1 means that the rod bundle interphase friction model will be applied, b = 2 means that the ORNL ANS interphase friction model will be applied (see card CCC0111 in Section 8.2.2).

f-flag The digit f specifies whether wall friction is to be computed. f = 0 specifies that wall friction effects are to be computed along the x-coordinate of the volume, and f = 1 specifies that wall friction effects are not to be computed along the x-coordinate.

e-flag The digit e specifies if nonequilibrium or equilibrium is to be used. e = 0 specifies that a nonequilibrium (unequal temperature) calculation is to be used, and e = 1 specifies that an equilibrium (equal temperature) calculation is to be used. Equilibrium volumes should not be connected to nonequilibrium volumes. The equilibrium option is provided only for comparison with other codes.

8.2.2 Single-Volume ORNL ANS Interphase Model Values, CCC0111

This card is required if the interphase friction flag b in Word 9 on cards CCC0101-0109 is set to 2.

W1(R) Gap (distance between side walls, short length, pitch, channel width) (m, ft).

W2(R) Span (distance from one end to the other, long length) (m, ft).

W3(I) Volume number.

8.2.3 Single-Volume Additional Wall Friction, CCC0131

This card is optional. If this card is not entered, the default values are 1.0 for the laminar shape factor and 0.0 for the viscosity ratio exponent. Two, four, or six quantities may be entered on the card, and the data not entered are set to default values.

W1(R) Shape factor for x-coordinate.

W2(R) Viscosity ratio exponent for x-coordinate.

W3(R) Shape factor for y-coordinate.

W4(R) Viscosity ratio exponent for y-coordinate.

W5(R) Shape factor for z-coordinate.

W6(R) Viscosity ratio exponent for z-coordinate.

8.2.4 Single-Volume Initial Conditions, CCC0200

This card is required for a single-volume.

W1(I) Control word. This word has the format εbt. It is not necessary to input leading zeros.

e-flag The digit ε specifies the fluid, where ε =0 is the default fluid. The value for ε > 0 corresponds to the position number of the fluid type indicated on the 120-9 cards (i.e., ε = 1 specifies H_2O, ε = 2 specifies D_2O, etc.). The default fluid is that set for the hydrodynamic system by cards 120-9 or this control word in another volume in this hydrodynamic system. The fluid type set on cards 120-9 or these control words must be consistent (i.e., not specify different fluids). If cards 120-9 are not entered and all control words use the default ε = 0, then H_2O is assumed as the fluid. It is recommended that the user use the 120-9 cards to set the fluid type in the systems.

b-flag The digit b specifies whether boron is present or not. The digit b = 0 specifies that the volume liquid does not contain boron; b = 1 specifies that a boron concentration in mass of boron per mass of liquid (which may be 0) is being entered after the other required thermodynamic information.

t-flag The digit t specifies how the following words are to be used to determine the initial thermodynamic state. Entering t = 0-3 specifies only one component (steam/water). Entering t = 4-6 allows the specification of two components (steam/water and noncondensable gas).

With options t equal to 4-6, names of the components of the noncondensable gas must be entered on card 110, and mass fractions of the noncondensable gas components are entered on card 115.

One Component (Steam/Water), Equilibrium or Non-equilibrium

[P,U_f,U_g,α_g] If t = 0, the next four words are interpreted as pressure (Pa, lb_f/in^2), liquid specific internal energy (J/kg, Btu/lb), vapor specific internal energy (J/kg, Btu/lb), and vapor void fraction. These quantities will be interpreted as nonequilibrium or equilibrium conditions depending on the internal energies used to define the thermodynamic state. Enter boron in W6 if needed.

One Component (Steam/Water), Equilibrium

[T,x_s] If t = 1, the next two words are interpreted as temperature (K, ^oF) and static quality in equilibrium condition. Enter boron in W4 if needed.

[P,x_s] If t = 2, the next two words are interpreted as pressure (Pa, lb_f/in^2) and static quality in equilibrium condition. Enter boron in W4 if needed.

[P,T] If t = 3, the next two words are interpreted as pressure (Pa, lb_f/in^2) and temperature (K, ^oF) in equilibrium condition. Enter boron in W4 if needed.

Two Components (Steam/Water/Air), Non-equilibrium

The following options are used for input of non-equilibrium two-phase and/or noncondensable states. In all cases, the criteria used for determining the range of values for static quality (x_s) are

Two-phase if $1.0E-9 \le x_s \le 0.99999999$,

Single phase if $x_s < 1.0$ E-9 or $x_s > 0.99999999$

The static quality is given by $x_s = M_g/(M_g + M_f)$, where $M_g = M_s + M_n$.

[P,T,x_s] If t = 4, the next three words are interpreted as pressure (Pa, lb_f/in^2), temperature (K, ^oF), and static quality in equilibrium condition. Using this input option with static quality > 0.0 and $\leq$ 1.0, saturated noncondensables will result. Also, the temperature is restricted to be less than the saturation temperature at the input pressure. Setting static quality to 0.0 is used as a flag that will initialize the volume to all noncondensable (dry noncondensable) with no temperature restrictions. Static quality is reset to 1.0 using this dry noncondensable option.

[T,x_s,x_n] If t = 5, the next three words are interpreted as steam saturation temperature (K, ^oF), static quality, and noncondensable quality in equilibrium condition. Both the static and noncondensable qualities are restricted to be between 1.0 E-9 and 0.99999999. The liquid will be subcooled. The input temperature determines the steam partial pressure, and the total pressure may be greater than the critical pressure, which will cause an input processing error. Little experience has been obtained using this option, and it has not been checked out.

[P,U_f,U_g,α_g,x_n] If t = 6, the next five words are interpreted as pressure (Pa, lb_f/in^2), liquid specific internal energy (J/kg, Btu/lb), vapor specific internal energy (J/kg, Btu/lb), vapor void fraction, and noncondensable quality. These quantities will be interpreted as nonequilibrium or equilibrium conditions depending on the internal energies used to define the thermodynamic state. The combinations of vapor void fraction and noncondensable quality must be thermodynamically consistent. If noncondensable quality is set to 0.0, noncondensables are not present and the input processing branches to that type of processing (t = 0). If noncondensables are present (noncondensable quality greater than 0.0), then the vapor void fraction must also be greater than 0.0. If the noncondensable quality is set to 1.0 (pure noncondensable), then the vapor void fraction must also be 1.0. When both the vapor void fraction and the noncondensable quality are set to 1.0, the volume temperature is calculated from the noncondensable energy equation using the input vapor specific internal energy.

W2-W7(R) Quantities as described under Word 1 (W1). Depending on the control word, 2-6 quantities may be required. Enter only the minimum number required. If entered, boron concentration (mass of boron per mass liquid) follows the last required word for thermodynamic conditions.

8.3 Time-Dependent Volume Component, TMDPVOL

This component is indicated by TMDPVOL for Word 2 on card CCC0000. For major edits, minor edits, and plot variables, the volume in the time-dependent volume component is numbered as CCC010000.